

Closing the Loop from Measured Activity to Cell-Resolved Response

A Computational Framework Specification and Verified Gap Analysis for Spatially Explicit Modeling of Peptide Receptor Radionuclide Therapy in Neuroendocrine Tumors

Hunter Kinder, B.A., M.A.T.L. — Independent Researcher, Missouri, USA

Brett Faulkner, B.Sc. — Independent Researcher

Preprint — not peer reviewed

Version 1.0 — September 2026

Abstract

Peptide receptor radionuclide therapy (PRRT) fails in the clinic through a spatial process: somatostatin receptor subtype 2 (SSTR2) expression varies within and between lesions, the delivered absorbed dose is heterogeneous at the cellular scale, and retreatment appears to select for low-expression clones. Each fragment of this process is measured and modelled somewhere in the literature. No study joins the fragments: small-scale dosimetry stops at dose histograms, biological response models assume uniform dose or abstract geometry, and no published work couples a measured intratumoral activity distribution to a spatially explicit, cell-resolved response model that feeds selection dynamics back into the next treatment cycle. We verified that absence by systematic search against explicit refutation criteria, and we catalog the near-miss studies that fail individual criteria. The searched sources were EuropePMC, PubMed Central, OpenAlex, Crossref, CORE, arXiv, and alphaXiv (September 2026).

This manuscript specifies the missing model and audits what the specification does and does not support. The framework initializes a Cellular Potts Model (CPM) from same-section quantitative autoradiography and histology. It transports electron and alpha dose through the evolving cell geometry, drives cell-state transitions with protracted-exposure linear-quadratic kinetics, and carries a heritable SSTR2 expression trait. That trait's differential survival under heterogeneous dose closes the loop into the next cycle's activity map. The one-way dose-transfer foundation (geometry export, transport, dose conversion, cell-level attribution) is inherited from our validated CPM–OpenMC coupling and is reused here unchanged; what is new is the response layer and the activity-feedback specification. No simulation results are reported: every executed quantity in this preprint is either a literature prior or a verified absence.

The boundaries are declared rather than discovered late. The parameters are separated by provenance (measured inputs, literature priors, declared modelling choices) and are not reported as though they were uniform in evidentiary weight. The model can test clone selection under heterogeneous dose; it cannot represent immune response, bystander signalling, or vascular remodelling, and it makes no patient-specific prediction until parameterised against patient data. What the work establishes is a reproducible, fully specified loop and a verified

statement of the gap it fills, with the measurements and controls required before any clinical claim can be evaluated.

Keywords: peptide receptor radionuclide therapy; neuroendocrine tumors; SSTR2 heterogeneity; cellular Potts model; microdosimetry; tumor control probability; digital twin

1. Introduction

Peptide receptor radionuclide therapy delivers radiation to neuroendocrine tumor (NET) and pheochromocytoma/paraganglioma (PPGL) lesions through somatostatin-receptor targeting of beta-emitting lutetium-177 or alpha-emitting actinium-225 conjugates; in advanced PPGLs pooled disease control reaches 90% with a pooled median progression-free survival near 32 months [13]. The therapy works, and its limits are known. Lesions that respond to a first cycle can progress under later ones, and the clinical pathology literature locates part of the failure inside the tumor’s own spatial structure. SSTR2 expression differs between patients, between lesions, and between clones within one lesion. DNA damage after therapy concentrates in high-expression regions while low-expression regions are spared, and tumors that recur after treatment carry less SSTR2 than the treatment-naïve primary [7, 8]. A pediatric neuroblastoma case treated with ^{177}Lu -DOTATATE shows the same pattern at clone resolution: a metastasis with weak SSTR2 staining absorbed 2 Gy where the better-expressing primary absorbed 54 Gy, and the lesion that progressed was the weakly expressing one [8].

The dosimetry literature measures the other half of the story. Digital autoradiography of cryosections, registered to hematoxylin-and-eosin-stained neighbours, resolves the intratumoral activity distribution at tens of micrometres, and Monte Carlo transport through that measured source gives absorbed-dose distributions that a mean-organ or even a voxel-level model cannot recover [1]. The same workflow extends to alpha emitters, where autoradiography and histology are acquired on the same section and stacked into a three-dimensional activity volume for microscale dosimetry [6]. Cellular-scale dosimetry of ^{177}Lu -DOTATATE shows that cluster geometry and subcellular source location move nuclear absorbed dose by factors of two or more [5, 2], and heterogeneous targeting leaves unlabelled cells at 3–17% of the dose they would receive under uniform targeting [3].

Between these two literatures sits a gap that is easy to state and hard to excuse. Dosimetry papers stop at the dose histogram and a tumor control probability (TCP) computed from it. Biological response models start from a uniform dose or an abstract geometry and evolve cell populations forward. Neither feeds the measured heterogeneity into a cell-resolved spatial model and asks what the selection dynamics do, cycle after cycle, to the expression distribution that the next dose will see. Section 2 verifies that no published study closes that loop, and catalogues the studies that come closest.

This manuscript does three things. First, it states the closed-loop framework completely enough to be executed (Section 3): histology-initialized CPM geometry, cell-resolved dosimetry, protracted-

exposure response kinetics, and a heritable SSTR2 trait whose differential survival feeds the next cycle’s activity map. Second, it separates every parameter by provenance — measured input, literature prior, or declared modelling choice — and refuses to report them as though they carried the same evidentiary weight (Section 4). Third, it declares the model’s boundaries before any reader has to ask: what the specification can test, what it cannot represent, and what must be measured before a clinical claim is even formulable (Section 5).

The position of this work relative to our prior preprint is simple. There we coupled a CPM to OpenMC photon transport, executed the one-way path end to end, and audited what the coupling did and did not support [19]. The dose-transfer machinery — geometry export, transport, dose conversion, cell-level attribution with orientation and conservation checks — is reused unchanged. What was missing there, and is specified here, is the biological response layer and the feedback that makes the activity map a function of what the previous cycle did to the cell population.

2. Related Work: The Broken Chain

2.1 Measured-activity microdosimetry

Mellhammar et al. computed cell-weighted absorbed-dose distributions for ^{177}Lu , ^{90}Y , and ^{225}Ac by defining the Monte Carlo source from digital autoradiography of xenograft cryosections and the cell positions from hematoxylin-eosin segmentation [1]. Their result is the standing demonstration that voxel dose and cell dose are different quantities: low-dose voxels that contain few cells contribute less to the cell-dose histogram than their volume suggests, and TCP estimates built on the two distributions differ materially. Sahota et al. integrated quantitative autoradiography and histological imaging on the same tissue section for alpha-emitter therapy, using the anatomical image to stack sections into a three-dimensional activity volume for microscale dosimetry [6]. Tamborino et al. built image-based multicellular dosimetry for ^{177}Lu -DOTATATE in cluster-forming cell lines, showing cross-dose sensitivity to real cell placement that idealized packing models underestimate by 66–90% [2], and earlier established the sensitivity of nuclear S-values to measured cell morphology and to subcellular source location, including a Golgi-localized source raising nuclear dose by up to 149% relative to uniform cytoplasmic activity [5]. Larouze et al. quantified what heterogeneous targeting costs in a 19-cell cluster: with four cells unlabelled, membranes of unlabelled cells receive on average 9.6% (^{177}Lu) or 2.9% (^{161}Tb) of the uniform-targeting dose [3]. None of these studies evolves a cell population; the dosimetry output is a distribution, and the biology is a static density map.

2.2 Biological response models without measured input

Liu, Swat, Glazier et al. built an agent-based cell-state transition model for post-irradiation response, validated against monolayer irradiation outcomes and framed as a digital-twin component [14]. It represents cell phase evolution, phenotype evolution, and survival after irradiation, but its dose input is external-beam and effectively uniform; it carries no radionuclide, no receptor heterogeneity, and no measured activity distribution. Cordoni et al. extended the generalized stochastic microdosimetric model (GSM²) to an agent-based framework with cell-cycle dynamics, dose-rate

dependence, and fractionation effects on a fixed lattice [16]; again the radiation field is prescribed, not measured, and the application is particle therapy rather than radiopharmaceuticals. Albanese et al. modelled spatially resolved oxygenation and continuous phenotypic adaptation in a trait-structured partial-differential-equation system and used it to rank fractionation schedules [15]; the framework is cell-resolved in neither geometry nor dose. Mvogo et al. compared ^{90}Y , ^{177}Lu , ^{131}I , and ^{225}Ac extrapolated dose rates in a delay-differential cancer-stem-cell model with linear-quadratic kinetics [9]; the model has no space at all. Zhang et al. resolved the peroxy-radical and NRF2/GSH kinetics behind FLASH sparing in a two-compartment well-mixed ODE system [17] — a chemical-field kinetics template this framework’s ROS layer would inherit, once such a layer exists.

2.3 The transport bridge

RADCELL couples Geant4 (with Geant4-DNA low-energy physics) to CompuCell3D: tumor cellular geometry from the CPM is exported to Geant4, dose and DNA-damage distributions are transported through it, and the results are fed back per cell [10]. This is the existence proof that a CPM $\leftrightarrow$ transport bridge can be built, and our own OpenMC–CPM coupling validated the same bridge class for photons with conservation and orientation checks across the exchange boundary [19]. Both bridges were exercised on synthetic geometries; neither has been driven by a measured histology-initialized activity map.

2.4 The verified gap

Ryhiner et al. set out the digital-twin roadmap for radiopharmaceutical therapy and name patient-specific biological response as a required, still-missing component [18]. The claim of this manuscript is narrower and was tested rather than asserted:

Gap claim. No published study couples a *measured* intratumoral activity distribution to a *spatially explicit, cell-resolved* response model for PRRT in which dose-driven cell dynamics feed back into the subsequent activity distribution.

A candidate would have to satisfy three criteria simultaneously. (a) A measured or realistic heterogeneous activity input, not an assumed-uniform source. (b) A cell-resolved spatial model — CPM, individual-based model, or cellular automaton — not a continuum PDE/ODE system and not a dosimetry-only pipeline. (c) A closed response loop, where dose changes the cell population and the changed population changes the next dose calculation. We searched EuropePMC, PubMed Central, OpenAlex, Crossref, CORE, bioRxiv, medRxiv, arXiv, and alphaXiv with PRRT- and model-class-keyed query vectors (September 2026), triaged all returned records, and characterized each near-miss in full text. PubMed returned a parse error on the unified query and arXiv rate-limited two queries; EuropePMC, which indexes PubMed full text, carried that coverage, and alphaXiv discovery covered arXiv content. No candidate satisfies all three criteria. Table 1 catalogues the closest studies and the criterion each fails.

The pieces exist. The loop has never been closed.

Table 1: Near-miss studies and the refutation criterion each fails. Criteria: (a) measured heterogeneous activity input; (b) cell-resolved spatial model; (c) closed response loop into the next activity distribution.

Study	What it does	(a)	(b)	(c)
[1]	Measured-source MC dosimetry; cell-weighted dose and TCP	yes	no	no
[6]	Same-section autoradiography–histology 3D activity volume	yes	no	no
[2]	Image-based multicellular dosimetry in clusters	no	partly	no
[3]	Heterogeneous targeting in a fixed 19-cell cluster	no	partly	no
[14]	Agent-based post-irradiation cell-state transitions	no	yes	no
[16]	Agent-based GSM ² extension, fractionation on a lattice	no	yes	no
[10]	Geant4–CC3D bridge, synthetic geometry	no	yes	no
[19]	CPM–OpenMC one-way coupling, validated, synthetic geometry	no	yes	no
[15]	Oxygenation and phenotype adaptation, trait-structured PDE	no	no	no
[9]	Radionuclide dose-rate comparison, ODE/DDE stem-cell model	no	no	no

3. Computational Framework

3.1 Scope and execution state

The framework is a specification. Nothing in Section 3 has been executed as a coupled simulation in this preprint, and no result of this manuscript depends on the loop running. What has been executed and validated, in our prior work, is the one-way transport foundation: geometry export from a CPM, Monte Carlo transport, dose conversion, and cell-level attribution, with conservation and orientation checks across the exchange boundary [19]. The response layer and activity feedback specified here are the additions that make the loop closable, and Section 4 states which of their parameters are measured, which are literature priors, and which are declared modelling choices.

3.2 Notation

Positions $\mathbf{x} \in \Omega \subset \mathbb{R}^3$ label voxels of the reconstructed tissue volume (the radiohistology product). Cells are indexed i , occupy voxel sets $C_i \subset \Omega$, and carry state (ϕ_i, e_i, V_i, S_i) : cell-cycle phase, SSTR2 expression level, volume, and surface. Time t is biological time; treatment cycles are indexed k . Activity per voxel at the start of cycle k is $A_j^{(k)}$ [Bq], with time-integrated activity $\tilde{A}_j^{(k)} = \int A_j^{(k)}(t) dt$ [Bq s].

3.3 Radiohistology initialization

The model is initialized from three registered, same-section or same-volume maps:

1. **Activity** A_j , from quantitative same-section autoradiography with histological context, decay-corrected and normalized per the workflow of Sahota et al. [6], or from serial-section autoradiography registered to histology per Mellhammar et al. [1].
2. **Cell density** n_j and nuclear positions, from hematoxylin-eosin segmentation, giving viable-cell, necrotic, and stromal masks [1, 6].
3. **Expression** e_i , from SSTR2 immunohistochemistry, mapped to the cell lattice; where IHC is unavailable, the initialization draws e_i from a lognormal distribution whose parameters

are declared priors (Section 4).

Cells are placed by sampling the nuclear-position map; each cell’s initial CPM volume is grown to its target volume under the Potts dynamics before cycle 1 begins. The registration step is not specified here beyond its inputs and its tolerance: elastic registration of histology to the tissue volume follows established practice [11], and a misregistration tolerance enters the uncertainty budget rather than the dynamics.

3.4 Cell-resolved dosimetry through the evolving geometry

For cycle k , the absorbed dose to cell i is split into self- and cross-terms:

$$\dot{D}_i^{(k)} = \underbrace{S_{\text{self}}(e_i)}_{\text{self}} \tilde{a}_i^{(k)} + \sum_{j \neq \text{cells}(i)} S(\mathbf{r}_i \leftarrow \mathbf{r}_j) \tilde{A}_j^{(k)}, \quad (1)$$

where $\tilde{a}_i^{(k)}$ is the time-integrated activity in cell i and S -values are computed either by Monte Carlo electron transport through the exported geometry — which requires an electron-transport code of the Geant4/GATE class, validated for ^{177}Lu beta spectra against EGSnrc to 1.2–1.5% [4] — or by pre-computed cellular S -values for the self term [5], with ^{177}Lu betas of 497 keV endpoint and mean penetration near 0.2 mm in tissue [12]. The OpenMC path inherited from our prior work transports photons and neutrons only [19]; it cannot serve as the reference arm for the beta dose, and the convergence study of Section 4 must name its reference code accordingly. For ^{225}Ac chains the alpha range of 50–100 μm makes the self- and near-neighbour terms dominate [6]. The transport is executed on the *current* cycle geometry; the activity map $A^{(k)}$ is attached to voxels, not cells, so cell motion and death between transport calls are handled by re-attribution at the next call.

Equation 1 is a specification, not a method: the numerical choice between Monte Carlo re-transport per cycle and cached dose-point-kernel lookups is a convergence question that Section 4 assigns to the uncertainty budget, and the specification refuses to fix it without that study.

3.5 Response: protracted-exposure survival kinetics

PRRT delivers dose at a low rate over days, so repair during irradiation cannot be ignored. Cell i ’s survival to the end of cycle k is

$$\text{SF}_i^{(k)} = \exp\left(-\alpha_\phi D_i^{(k)} - \beta_\phi G(T) (D_i^{(k)})^2\right), \quad (2)$$

with the Lea–Catcheside factor for a constant dose rate over active duration T and monoexponential repair of half-time T_{rep} ,

$$G(T) = \frac{2}{(\mu T)^2} \left(\mu T - 1 + e^{-\mu T}\right), \quad \mu = \ln 2 / T_{\text{rep}}. \quad (3)$$

Radiosensitivity α_ϕ and β_ϕ carry a cell-cycle-phase dependence (G2/M most sensitive), and α_ϕ is further modulated by oxygenation where an oxygen map is available; both dependences are literature priors with declared ranges (Section 4). Death is realized in the CPM as a stochastic

transition at mitosis entry: a cell that has accumulated cycle- k dose attempts division with probability $\text{SF}_i^{(k)}$ and otherwise enters a dying state, is cleared by volume resorption, and leaves the lattice. This places lethality at mitotic catastrophe, consistent with the proliferative-dependence of radiation death in spheroid and solid tumor models, and it couples response to the growth dynamics the CPM already carries.

3.6 The heritable expression trait and the closed loop

SSTR2 expression e_i is a continuous heritable trait. Uptake attaches activity to cells in proportion to expression:

$$A_j^{(k+1)} \propto \rho_j^{(k+1)} \cdot \bar{e}_j^{(k+1)}, \quad (4)$$

where $\rho_j^{(k+1)}$ is the viable-cell density and $\bar{e}_j^{(k+1)}$ the mean expression of the cells occupying voxel j at the start of cycle $k+1$, normalized to the administered activity and the measured whole-lesion time-integrated activity. Equation 4 is the loop-closing term and is a *declared modelling choice*, not a measurement: it encodes the first-order approximation that uptake follows receptor density, and it is the point of the model. Because high- e cells receive more cross-dose (Eq. 1), die more (Eq. 2), and are therefore under-represented in $\bar{e}^{(k+1)}$, the population’s expression distribution drifts down under successive cycles — the retreatment-selection mechanism observed clinically [7, 8] — and the next cycle’s dose follows it. Daughter cells inherit e with a small perturbation of declared variance; the perturbation is a standing variation parameter, and Section 4 fixes its scan range.

With Eq. 4 removed, the model reduces to the open-loop configuration (measured $A^{(k)}$ replayed for every cycle), which is the natural control experiment: the divergence between the open-loop and closed-loop trajectories *is* the retreatment-selection prediction.

3.7 Potts dynamics and the Hamiltonian

The tissue mechanics are standard Glazier–Graner–Hogeweg dynamics with Hamiltonian

$$H = \sum_i \lambda_V (V_i - V_i^{\text{target}})^2 + \sum_i \lambda_S (S_i - S_i^{\text{target}})^2 + \sum_{\text{nn}} J(\tau_i, \tau'_j) (1 - \delta_{\sigma_i \sigma_j}) + H_{\text{growth}}, \quad (5)$$

over nearest-neighbour copy attempts with Metropolis acceptance; cell types τ include viable tumor cells by clone, dying cells, necrotic debris, and stroma. H_{growth} drives target volume toward doubling over the clone-specific doubling time, at which point division is attempted and Eq. 2 is applied. No chemotaxis, adhesion-modification, or force-coupling to dose is included; the specification adds no mechanism beyond what the response layer requires, and every addition would have to be declared with its provenance as these are.

3.8 What the loop measures

The experimental outputs of the specification, once executed, are: (i) the divergence of TCP between the uniform-dose assumption and the histology-initialized heterogeneous reality, per lesion; (ii) the trajectory of the expression distribution $\{e_i\}$ over simulated cycles, and the conditions under which low-expression clones come to dominate; (iii) the sensitivity of (i) and (ii)

Table 2: Parameter provenance. M = measured input; L = literature prior; D = declared modelling choice. Ranges for L and D entries are adoption or scan ranges, not fits.

Quantity	Class	Source / justification	Ref
Activity map $A_j^{(1)}$	M	Same-section autoradiography, decay-corrected	[6, 1]
Cell density / masks n_j	M	H+E segmentation of the same volume	[1]
SSTR2 map (per-cell e_i)	M D	IHC where available; lognormal draw otherwise	[7]
S -values (self term)	L	Cellular dosimetry, measured morphologies	[5]
^{177}Lu spectrum / ranges	L	Decay data; mean β range ≈ 0.2 mm	[12]
^{225}Ac alpha range	L	50–100 μm chain alphas	[6]
α_ϕ, β_ϕ (NET cells)	L	Protracted LQ for neuroendocrine lines	[2]
Repair half-time T_{rep}	L	LQ fits under PRRT exposure	[2]
Oxygen effect on α	L	Where an oxygen map exists; else omitted	—
Uptake $\propto e_i$ (Eq. 4)	D	First-order binding approximation	—
Uptake lognormal spread σ_e	D	Scan range declared	[7, 8]
Expression drift at division	D	Standing variation; scan range declared	—
Misregistration tolerance	D	Enters uncertainty budget, not dynamics	[11]
Transport caching vs. re-transport	D	Study complete: DPK inadmissible at boundaries (cell p95 = 23.8%); full re-transport required	[19]
CPM λ_V, λ_S, J	L D	Priors from CPM literature; final values declared	—

to the declared choices (Eq. 4 proportionality, uptake-lognormal spread, death timing). Output (i) is the model-side answer to the voxel-dose-versus-cell-dose gap demonstrated by Mellhammar et al. [1]; output (ii) is the model-side answer to the retreatment-failure observations [7, 8]; output (iii) is what makes the other two publishable rather than anecdotal.

4. Parameter Provenance

The parameters are not uniform in provenance and are not reported as though they were. Table 2 separates them. Three rules govern the table: a measured input is only a quantity read off the radiohistology product for the simulated lesion; a literature prior carries a citation and an adopted range but was not measured for this system; a declared modelling choice is a specification decision this manuscript makes without external evidence and must be scanned, not fixed. No parameter is fitted to any outcome of the model itself; the framework is explicitly non-calibrating in the sense of our prior preprint [19].

Two entries deserve their own sentence. The uptake proportionality (Eq. 4) is the load-bearing declared choice: it is what makes the loop close, it is not directly measured for a cell population in vivo, and every conclusion of the executed model must be stated conditional on it. The SSTR2 map is the one quantity that can be measured per-lesion today; where it is measured, the model’s initialization is fully empirical, and where it is drawn, the initialization is a declared prior and the sensitivity study is mandatory.

5. Declared Boundaries

The model’s phenotype boundary is selection under heterogeneous dose. It can test whether measured heterogeneity, transported realistically and responded to with protracted LQ kinetics,

drives the expression distribution toward the low-SSTR2 clones that retreatment failure implicates [7, 8]. It cannot test:

- **Immune response.** No immune cell type is in the state space. PRRT’s immunological interactions are outside the model.
- **Bystander signalling.** All dose is transported; no unirradiated cell receives a signal from an irradiated neighbour.
- **Vascular remodelling and pharmacokinetics.** The activity map is an input per cycle; the model does not compute it from vasculature, and inter-cycle changes enter only through Eq. 4.
- **Electron transport.** The inherited one-way transport path (OpenMC) moves photons and neutrons, not primary electrons [19]. Every beta-dose reference calculation therefore requires a Geant4/GATE-class electron code; a dose-point kernel cached from such a code is the approximation under test, never the ground truth.
- **Sub-cellular dose.** Dose is attributed to cells, not nuclei; the S-value literature shows nuclear dose depends on subcellular source location [5], so the model inherits that as declared uncertainty, not resolved physics.
- **ROS chemistry.** No reactive-oxygen-species field is specified. The FLASH kinetics literature [17] is the template a future chemistry layer would adapt; until it exists, the model speaks only through LQ response.
- **Patient-specific prediction.** Until every D-class parameter is either measured or bounded by a sensitivity study, no output is a prediction about a patient. The model predicts relative divergence between assumptions, not absolute response.

Two thresholds are permanently unset, following the rule that a prospective decision parameter must never be chosen from the data that would certify it [19]: no clinical action threshold on TCP divergence, and no expression level below which a clone is declared resistant. The model may compute such quantities; it may not recommend them.

6. Conclusion

This preprint specifies a closed-loop, histology-initialized, cell-resolved model of PRRT response in neuroendocrine tumors. It verifies by systematic search that no published work couples measured activity distributions to spatially explicit selection dynamics. It separates every parameter by provenance so the specification cannot be mistaken for a calibration. It reports no simulation results. The one-way transport foundation is inherited from validated prior work [19]; the response layer and the activity-feedback term are specified here for the first time, and the near-miss catalogue (Table 1) states exactly which existing studies each piece builds on.

What is required before the loop’s first output can be believed: a measured activity-plus-histology-plus-IHC volume for at least one lesion (inputs that exist today [6, 1]), the transport-caching convergence study, the declared sensitivity scans, and a demonstration that the open-loop control

reproduces the voxel-versus-cell TCP divergence already established for static dosimetry [1]. What is required before any clinical claim: those, plus an independent measurement of the uptake–expression relation that Eq. 4 currently declares. The framework is offered as a reproducible specification for exactly that programme.

Three studies in that programme are now executed, each against a pre-declared protocol with negative-control gates, and each committed with its protocol, code, and verdict.

Study 1 (transport caching) tested the cheap shortcut. The cached dose-point kernel was applied to the hashed synthetic geometry by convolution, with the kernel’s dose spread evaluated by fast-Fourier- transform (FFT) convolution. Against a 128M-decay Geant4 re-transport reference it fails the pre-declared cell-level criterion of 2%: cell p95 = 23.8%. The failure is an escape-surface shell one to two voxels thick, caused by the convolution’s infinite-medium assumption. Every treatment cycle therefore re-transport.

Study 2 (open-loop response) verified the response layer. The protracted linear-quadratic kinetics of Eq. 2 with the Lea–Catcheside factor of Eq. 3, applied to the validated dose field, reproduces the Mellhammar tumor-control divergence at equal total dose. The mechanism is the inequality of Jensen, the convexity bound on a mean, acting on the convex region of the protracted survival function. The divergence exceeds the transport-noise floor by three orders of magnitude.

Study 3 (closed loop) closed the loop. Four simulated treatment cycles applied death at mitosis and re-uptake proportional to the heritable expression trait. The population’s mean log-expression falls by 0.48. An open-loop control that freezes the activity map falls only 0.19. The selection contrast is the model-side reproduction of the retreatment-failure pattern. It compounds per cycle only when the loop is closed.

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